

Power BI Measures Documentation

This document contains the DAX measures used in the Manufacturing Performance Dashboard project. Each measure includes its formula and a brief explanation of its business purpose.

AVG Downtime

DAX Formula:

`DIVIDE([Total Downtime (Min)],[Total Batches],0)`

Description:

Calculates the average downtime per batch by dividing total downtime minutes by total number of batches.

AVG Ideal Production Time (Min)

DAX Formula:

`AVERAGEX(Fact_Production,RELATED(Dim_Products[Min batch time]))`

Description:

Calculates the average ideal production time based on the standard batch time defined for each product.

AVG Production Time (Min)

DAX Formula:

`AVERAGE(Fact_Production[Actual Batch Time])`

Description:

Calculates the average actual production time across all batches.

Total Batches

DAX Formula:

`DISTINCTCOUNT(Fact_Production[Batch])`

Description:

Counts the total number of unique batches produced.

On_Time Batches

DAX Formula:

`SUMX(
Fact_Production,
if(Fact_Production[Actual Batch Time]<= RELATED(Dim_Products[Min batch time]),1,0)
)`

Description:

Counts batches completed within or below the ideal production time.

On_Time Batches %

DAX Formula:

`DIVIDE([On_Time Batches],[Total Batches],0)`

Description:

Calculates the percentage of batches completed on time.

Delayed Batches

DAX Formula:

`[Total Batches]-[On_Time Batches]`

Description:

Calculates the number of delayed batches by subtracting on-time batches from total batches.

Delayed Batches %**DAX Formula:**

DIVIDE([Delayed Batches],[Total Batches],0)

Description:

Calculates the percentage of delayed batches out of total batches.

Total Downtime (Min)**DAX Formula:**

SUM(Dim_Downtime[Downtime Minutes])

Description:

Calculates the total downtime minutes recorded.

Total Actual Production Time (Min)**DAX Formula:**

SUM(Fact_Production[Actual Batch Time])

Description:

Calculates the total actual production time across all batches.

Downtime %**DAX Formula:**

DIVIDE([Total Downtime (Min)],[Total Actual Production Time (Min)],0)

Description:

Calculates downtime as a percentage of total actual production time.

Total Ideal Time (Min)**DAX Formula:**

SUMX(Fact_Production,RELATED(Dim_Products[Min batch time]))

Description:

Calculates the total ideal production time based on product standards.

Efficiency % (Based on Time)**DAX Formula:**

DIVIDE([Total Ideal Time (Min)],[Total Actual Production Time (Min)],0)

Description:

Measures production efficiency by comparing ideal production time with actual production time.

Expected batch efficiency Narrative**DAX Formula:**

var days=ROUND([Production minutes per day]*[Efficiency % (Based on Time)]/[AVG Production Time (Min)],0)
RETURN "Predicted Batches for next day is " & days& " batches based on previous EFFICIENCY"

Description:

Predicts the number of batches expected for the next day based on previous production efficiency.

Expected batches rolling historical average

DAX Formula:

var expectedDays=

```
AVERAGEX(  
    TOPN(3,  
        ALL(Fact_Production[Date]),Fact_Production[Date], DESC),  
    CALCULATE(DISTINCTCOUNT(Fact_Production[Batch])  
    )  
)  
  
RETURN
```

"Predicted Downtime for next day is " & ROUND(expectedDays,0) & " based on Rolling Historical Average for last 3 days Disadvantage for this method it does not consider DOWNTIME"

Description:

Predicts the expected number of batches using the rolling historical average of the last 3 days.

Factor Contribution %

DAX Formula:

```
DIVIDE([Total Downtime (Min)],CALCULATE([Total Downtime (Min)],REMOVEFILTERS(Dim_Downtime)))*100
```

Description:

Calculates the contribution percentage of each downtime factor relative to total downtime.

Human Downtime

DAX Formula:

```
CALCULATE([Total Downtime (Min)],Dim_Downtime_Factors[Operator Error] = TRUE())
```

Description:

Calculates total downtime caused by human/operator-related issues.

Human Downtime %

DAX Formula:

```
DIVIDE([Human Downtime], [Total Downtime (Min)])
```

Description:

Calculates the percentage of downtime caused by human factors.

Ideal Production Time %

DAX Formula:

```
DIVIDE([Total Ideal Time (Min)],[Total Actual Production Time (Min)],0)
```

Description:

Calculates the ratio of ideal production time to actual production time.

Predicted Downtime Narrative

DAX Formula:

var PredictedDowntime=

```

AVERAGEX(
    TOPN(3,
        ALL(Fact_Production[Date]),Fact_Production[Date], DESC
    ),
    CALCULATE(SUM(Dim_Downtime[Downtime Minutes])
)
)
RETURN
    "Predicted Downtime for next day is " & ROUND(PredictedDowntime,0) & " based on Rolling Historical
    Average for last 3 days"

```

Description:

Predicts the expected downtime for the next operational day using the rolling average of the last 3 days.

System Downtime

DAX Formula:

```
CALCULATE([Total Downtime (Min)],Dim_Downtime_Factors[Operator Error] = FALSE())
```

Description:

Calculates total downtime caused by system or machine-related issues.

System Downtime %

DAX Formula:

```
DIVIDE([System Downtime], [Total Downtime (Min)])
```

Description:

Calculates the percentage of downtime caused by system-related issues.

Top Factor Rank

DAX Formula:

```
RANKX(ALL(Dim_Downtime_Factors[Description]),[Total Downtime (Min)],,DESC)
```

Description:

Ranks downtime factors from highest to lowest based on total downtime minutes.

Total Operators

DAX Formula:

```
DISTINCTCOUNT(Fact_Production[Operator])
```

Description:

Counts the total number of unique operators involved in production.

Total Products

DAX Formula:

```
DISTINCTCOUNT(Fact_Production[Product])
```

Description:

Counts the total number of unique products produced.